Lakshya Jindal

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Education

International Institute of Information Technology, Hyderabad — Aug 2023 – Present
B.Tech + MS(By Research) in Electronics and Communication Engineering — *Hyderabad, India*

Hiranandani Foundation School Thane — June 2021 – June 2023
ISC, PCM and Computer Applications - 94.25% — *Mumbai, India*

Experience

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad — April 2025 – Present
Research Assistant — *On-site*

- Designed and implemented scalable motion planning algorithms in C++ and Python, improving performance through iterative optimization, testing, and debugging.
- Built modular software pipelines for large-scale algorithm evaluation, improving reproducibility, reliability, and maintainability.
- Collaborated with researchers to develop production-quality software from research prototypes, participating across design, implementation, evaluation, and continuous iteration.

Projects

Real-Time Chat Application | *React.js, Node.js, Express.js, MongoDB, Socket.IO, JWT, Tailwind CSS* | GitHub

- Designed and developed a full-stack real-time messaging application with secure **JWT authentication**, REST APIs, and MongoDB for persistent data storage.
- Implemented low-latency communication using Socket.IO for a distributed real-time messaging system supporting presence tracking, delivery acknowledgements, and read receipts
- Owned the complete software development lifecycle including design, implementation, testing, debugging, and deployment using Vercel, Render, and MongoDB Atlas.

Custom C-Shell | *C, POSIX APIs, Linux, Socket Programming*

- Built a Unix-like shell in C supporting command execution, process creation, piping, and input/output redirection using POSIX system calls.
- Implemented process management, signal handling, networking utilities, and persistent command history while improving modularity and maintainability of the codebase.

RISC-V Processor Design | *SystemVerilog, Computer Architecture* | GitHub

- Designed sequential and 5-stage pipelined RISC-V processors implementing hazard detection, forwarding, and control logic to improve execution efficiency.
- Verified processor correctness through simulation and comprehensive testing across control hazards, data hazards, and instruction execution scenarios.

LiDAR-Camera Calibration | *Python, Computer Vision, Sensor Fusion, Optimization* | GitHub

- Implemented an end-to-end LiDAR-camera calibration pipeline from the MDPCalib paper using Python and optimization techniques.
- Developed modular components for feature extraction, correspondence matching, and nonlinear optimization to accurately estimate sensor extrinsics.

NaviSight – Edge AI Wearable Device for Visually Impaired | *Computer Vision, Edge AI* | Website

- Fine-tuned a YOLOv11 model for real-world object detection, improving inference accuracy for assistive navigation.
- Built an offline inference pipeline optimized for latency and resource-constrained edge devices.

Technical Skills

Languages: C, C++, Python, SQL, Shell
Frameworks/Technologies: Linux, POSIX, React.js, Node.js, Express.js, MongoDB, Socket.IO, REST APIs
Developer Tools: Git, GitHub, Linux, ROS, MATLAB, Gazebo, Arduino

Achievements/Extracurriculars

- Achieved a peak rating of **1269** on **Codeforces**
- Secured **3rd Place** in **Megathon 2025** , Built Multi-Modal AI Solution for Estimating Damage Claim for Cars.
- Solved 200+ Data Structures and Algorithms problems across Codeforces, LeetCode and CodeChef.
- Secured **AIR 4391** in JEE Mains 2023 and **AIR 10931** in JEE Advanced 2023
- Cleared **KVPY 2022** and **Pre-RMO 2019**

Relevant Coursework

Data Structures and Algorithms | Operating Systems and Networks | Database Systems | Probability & Random Processes | Signal Processing | Robotics: Planning and Navigation | Intro to Processor Architecture |
| Computer Vision | Statistical Methods in AI | Communication Theory | Mechatronics System Design |
| Introduction to IoT | Object-Oriented Programming | Mobile Robotics | Data Analytics I | Data & Applications |